

Task 1: 8-Bit Arithmetic Logic Unit (ALU) Design & Verification

1. Project Overview

This project presents the Verilog HDL implementation, functional simulation, synthesis, and gate-level netlist analysis of an 8-bit Arithmetic Logic Unit (ALU) using AMD Xilinx Vivado. The design handles fundamental arithmetic operations, bitwise logic operations, bit shifts, and dynamic flag evaluation.

2. RTL Source Code (alu_8bit.v)

Verilog

```
`timescale 1ns/1ps
```

```
module alu_8bit (
```

```
    input [7:0] A,
```

```
    input [7:0] B,
```

```
    input [2:0] opcode,
```

```
    output reg [7:0] result,
```

```
    output reg zero_flag
```

```
);
```

```
always @(*) begin
```

```
    case (opcode)
```

```
        3'b000: result = A + B;    // Addition
```

```
        3'b001: result = A - B;    // Subtraction
```

```
        3'b010: result = A & B;    // Bitwise AND
```

```
        3'b011: result = A | B;    // Bitwise OR
```

```
        3'b100: result = A ^ B;    // Bitwise XOR
```

```
        3'b101: result = A << 1;    // Logical Shift Left
```

```
        3'b110: result = A >> 1;    // Logical Shift Right
```

```
        default: result = 8'h00;
```

```
    endcase
```

```
    // Zero Flag Evaluation
```

```
    zero_flag = (result == 8'h00) ? 1'b1 : 1'b0;
```

```
end
```

```
endmodule
```

3. Testbench Source Code (tb_alu_8bit.v)

Verilog

```
`timescale 1ns/1ps
```

```
module tb_alu_8bit;
```

```
    reg [7:0] A;
```

```
    reg [7:0] B;
```

```
    reg [2:0] opcode;
```

```
    wire [7:0] result;
```

```
    wire zero_flag;
```

```
    // Instantiate Unit Under Test (UUT)
```

```
    alu_8bit uut (
```

```
        .A(A),
```

```
        .B(B),
```

```
        .opcode(opcode),
```

```
        .result(result),
```

```
        .zero_flag(zero_flag)
```

```
    );
```

```
    integer i;
```

```
    initial begin
```

```
        // Directed Tests
```

```
        A = 8'h05; B = 8'h03; opcode = 3'b000; #10; // ADD: 5 + 3 = 8
```

```
        A = 8'h0A; B = 8'h04; opcode = 3'b001; #10; // SUB: 10 - 4 = 6
```

```
        A = 8'hFF; B = 8'h0F; opcode = 3'b010; #10; // AND: 0xFF & 0x0F = 0x0F
```

```
        // Random Tests
```

```
        for (i = 0; i < 5; i = i + 1) begin
```

```
            A = $random;
```

```
            B = $random;
```

```
opcode = $random % 7;
#10;
end
```

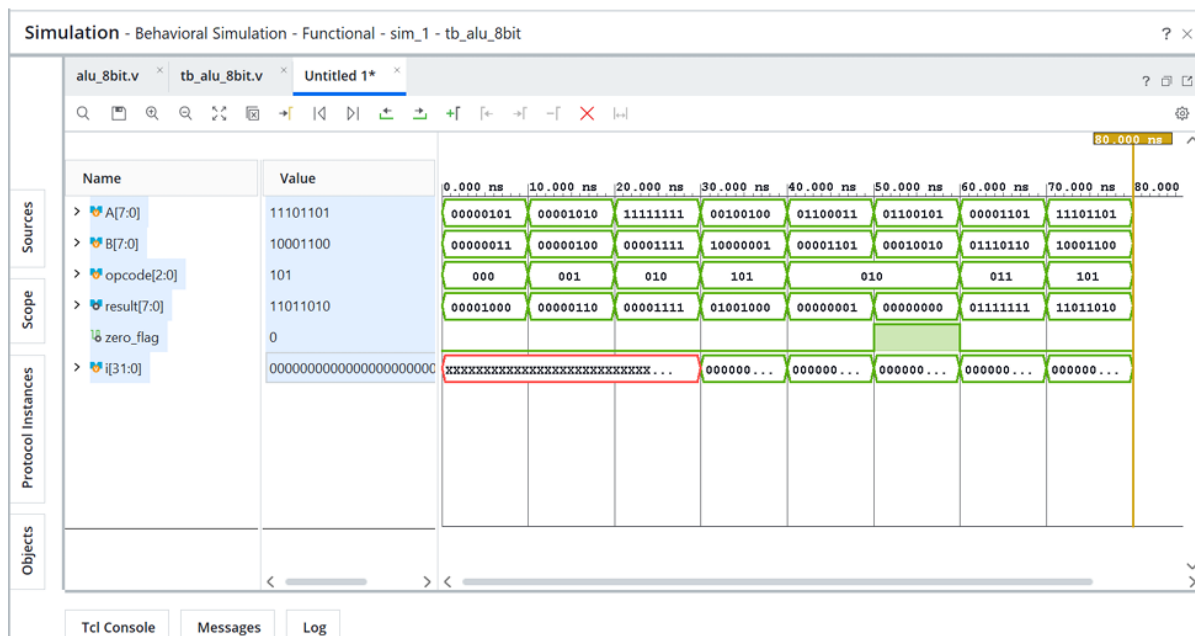
\$finish;

end

endmodule

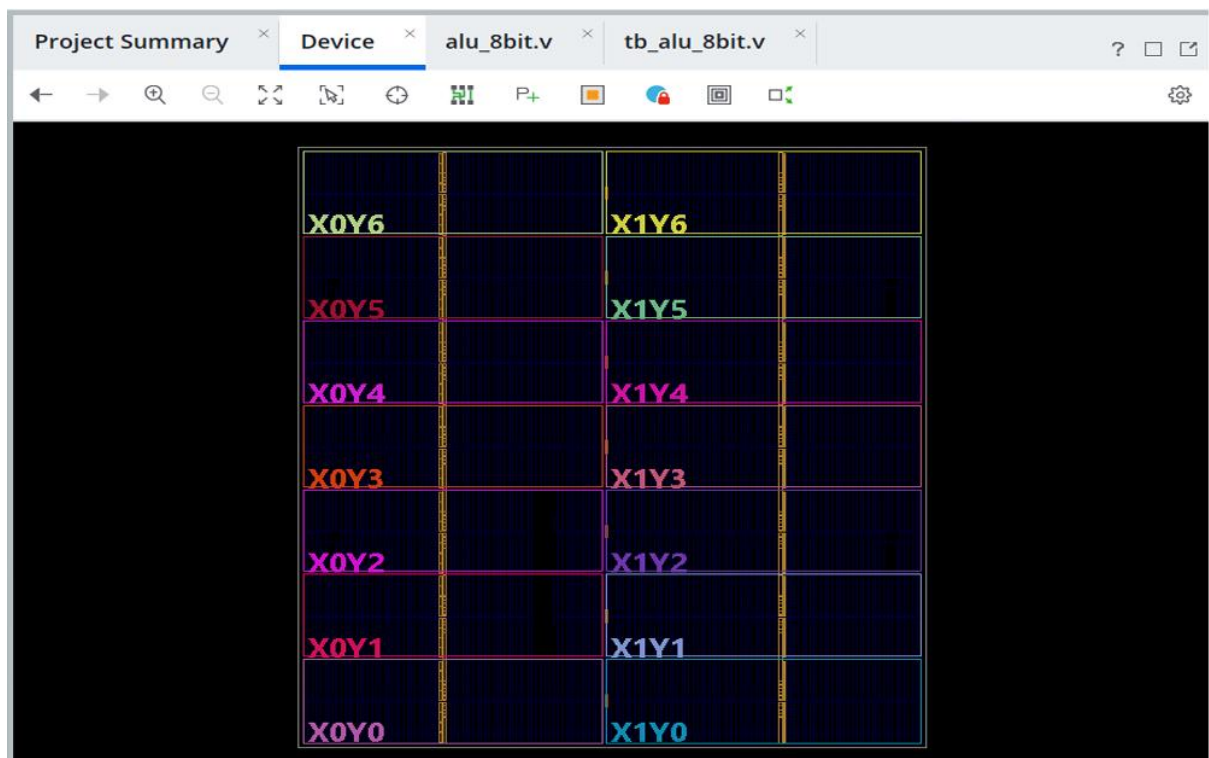
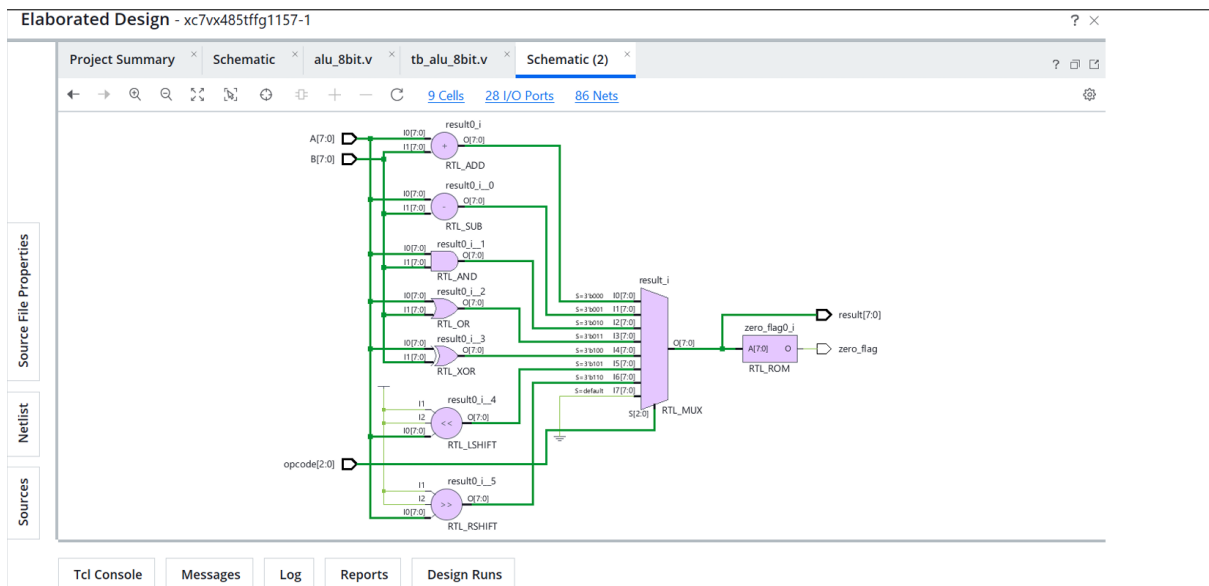
4. Deliverable Screenshots & Descriptions

A. Behavioral Simulation Waveform



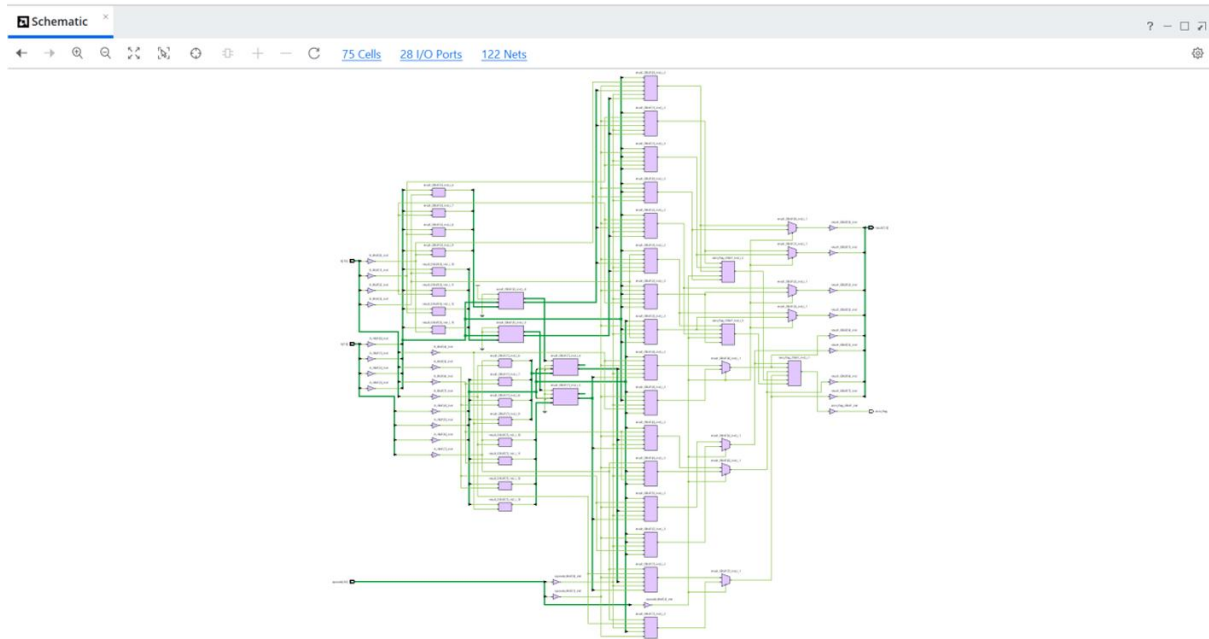
Analysis: The simulation demonstrates complete functional correctness. Directed tests verify addition ($5 + 3 = 8$), subtraction ($10 - 4 = 6$), and logic masking. Randomized test cases verify dynamic operation switching and proper assertion of zero flag at timestamp 50 ns.

B. Elaborated RTL Block Schematic



Analysis: Shows top-level functional RTL blocks generated by Vivado, including RTL_ADD, RTL_SUB, RTL_AND, RTL_OR, RTL_XOR, shift modules, and the central multiplexer (RTL_MUX).

C. Synthesized Gate-Level Netlist Schematic



Analysis: Illustrates the low-level mapping of the design into 75 FPGA hardware cells, primitive buffers (IBUF, OBUF), and lookup table blocks (LUT6) mapped to the target architecture.

5. Synthesis & Resource Utilization Summary

- **Target Device:** AMD Xilinx Virtex-7 (xc7vx485tffg1157-1)
- **Total Cells:** 75
- **I/O Ports:** 28 (19 Inputs, 9 Outputs)
- **Total Nets:** 122
- **Combinational Logic Usage:** The 8-bit combinational ALU synthesizes cleanly into hardware utilizing lookup tables (LUTs) without requiring flip-flops or sequential memory registers.